# 28Sep Lecture8

Monday, 28 September 2026 11:35 AM

27 September 2026.

Deadline → End October or November.

CUES / QUESTIONS

MARK WILL BE AWARDED IN FORM OF ASSINGMENT 3 OR 4.

10 Marks

* link of Excel Sheet for group will be shared Soon.

* we can also make project individualy.

* we have to make team our self.

NOTES MANDATORY PROJECT

## 1. Project Overview

Industry / Domain: [Healthcare / Banking / Manufacturing / Retail / Education / Logistics / Other]
Use Case: [One-sentence description of the application problem]
Business Need: Why does the industry need this system?
Target Users: [Customers / Operators / Administrators / Analysts]
Cloud Objective: Deploy a scalable, reliable and measurable application pipeline.
Team Members: [Names, Roll Numbers] (3 member) (maximum 5, minimum 3)

## 2. Industry Problem & Proposed Solution

Current Process / Pain Point: [Describe existing process and limitations]
Problem Statement: [Specific technical/business problem]
Proposed Solution: [Cloud-enabled application pipeline]
Expected Improvement: Availability, response time, scalability, automation and operational efficiency.
Key Assumptions & Constraints: [Data volume, users, budget, compliance, latency, etc.]

## 3. SLA & TTR Definition

SLA = Service Level Agreement | TTR = Time to Recovery / Resolution

Availability SLA: [e.g., 99.5% / 99.9%]
Application Response SLA: [e.g., < 2 seconds for 95% requests]
Recovery Objective: [RTO = ___ minutes]
Data Loss Objective: [RPO = ___ minutes]
TTR Target: [___ minutes from incident detection to service restoration]
Incident Severity: S1 Critical | S2 High | S3 Medium | S4 Low
Define how SLA violations will be detected and reported.

SUMMARY

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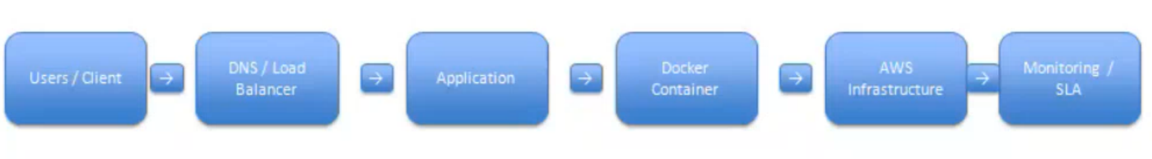
## 4. End-to-End Cloud Pipeline

Add diagram in project



Show the complete request-to-response and operations flow.
Mark where SLA metrics, logging, monitoring and failure recovery are implemented.

## 5. AWS Services Used

Compute: [EC2 / ECS / EKS / Lambda]
Storage: [S3 / EBS / EFS]
Database: [RDS / DynamoDB / Other]
Networking: [VPC / Security Groups / Load Balancer / Route 53]
Monitoring: [CloudWatch / CloudTrail]
Security: [IAM / Secrets Manager / KMS]
Deployment / Automation: [ECR / CodePipeline / CodeBuild / Terraform, if used]
For each service: state WHY it was selected and what role it plays.

we have to include Everything in the project.

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## 6. Docker & Containerization

Application Components: [Frontend / Backend / API / Database / Worker]
Docker Image: [Image name / version]
Dockerfile: Base image, dependencies, exposed port and startup command
Container Networking: [Bridge / Compose network / Cloud networking]
Persistent Data: [Volume / AWS managed storage]
Image Registry: [Amazon ECR or other]
Explain portability, isolation and reproducibility benefits.

## 7. Virtualization & Cloud Infrastructure

Virtualization Layer: [VM / Hypervisor / Container runtime]
VM Configuration: vCPU = ___ | RAM = ___ GB | Storage = ___ GB
Operating System: [Linux / Windows]
Container vs VM: Explain where each is used and why.
Resource Allocation: CPU, memory, network and storage requirements.
Scalability Strategy: Vertical scaling / Horizontal scaling / Auto Scaling.

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## 8. Reliability, Monitoring & TTR Workflow

Detection: [CloudWatch metric / log / health check / alert]
Incident Trigger: [Example failure condition]
Diagnosis: [Logs, metrics, traces, health checks]
Recovery Action: [Restart container / replace instance / rollback / scale-out]
Verification: [Health check + application test]
TTR Measurement = Detection Time + Diagnosis Time + Recovery Time + Verification Time
Target TTR: ___ minutes | Measured TTR: ___ minutes

## 9. Performance & SLA Testing

Test Workload: [Concurrent users / requests per second / data volume]
Latency: Average ___ ms | P95 ___ ms | P99 ___ ms
Throughput: ___ requests/sec
Availability Observed: ___ %
Failure Injection: [Container crash / instance failure / network interruption]
TTR Observed: ___ minutes
SLA Result: [Met / Partially Met / Not Met — supported by measurements]

we will get marks for this project.

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## 10. Security & Cost Considerations

IAM: Least-privilege access and role-based permissions
Network Security: VPC, security groups, restricted ports
Secrets: No credentials hard-coded in source code or Docker images
Data Protection: Encryption at rest / in transit
Cost Model: Compute + Storage + Database + Network + Monitoring
Estimated Monthly Cost: ₹_______ / month
Optimization: Right-sizing, auto-scaling, scheduled shutdown, managed services.

Add in Git hub, with sharable link.

Add proofs that I have made it myself.

## 11. Demonstration / Deployment Evidence

GitHub / Repository: [URL]
Docker Build & Run: [Command / screenshot]
AWS Deployment: [Screenshot / architecture]
Application URL / Endpoint: [URL]
CloudWatch / Monitoring Dashboard: [Screenshot]
Failure & Recovery Demonstration: [Before → Failure → Recovery]
Include 2–4 screenshots or a short live demonstration.

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## 12. Results & Key Findings

Functional Outcome: What was successfully implemented?
SLA Achievement: Target vs Observed
TTR Achievement: Target vs Observed
Performance: Key latency/throughput results
Reliability: Failure scenarios successfully handled
Cloud Benefits: Scalability, availability, automation, operational visibility
Limitations: [What remains to be improved?]

### 13. Conclusion & Future Work

Conclusion: Summarize the complete cloud application pipeline in 2–3 points.
Future Work: Kubernetes/EKS, CI/CD, Infrastructure as Code, serverless components, advanced observability, multi-region deployment.
Business Extension: How can the prototype become an industry-ready solution?
Learning Outcomes: AWS + Docker + virtualization + SLA/TTR + cloud deployment.

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## 14. Evaluation Checklist

- ☐ Clear industry use case and problem statement
- ☐ End-to-end cloud architecture
- ☐ At least 3–5 relevant AWS services with justification
- ☐ Dockerized application
- ☐ Virtualization / VM / container infrastructure explained
- ☐ Explicit SLA metrics and measurable targets
- ☐ TTR defined and experimentally measured
- ☐ Failure/recovery scenario demonstrated
- ☐ Monitoring/logging evidence
- ☐ Performance, security and cost discussion
- ☐ Working demo + repository/documentation

SUMMARY